

Mobile Communication Project Report: Simulation of MIMO OFDM System using Matlab

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Abstract

This paper details the theories and simulations of a 2×2 MIMO OFDM system. The simulation is built using Matlab and the Communication Toolbox plugin. By altering the parameters of the program, we observe the effects of each component to the whole system.

1 Introduction

Nowadays, with the developments in technology, the problem of fast data transmission has become increasingly relevant. New inventions built on the modern communication system such as smartphones, 5G, advanced Wi-Fi standards, etc. expect a fast and reliable method of transmitting and receiving information.

It is well known that a transmitting wave may be modified by the properties of the medium in which the wave propagates. These effects include attenuation, phase and frequency shifting

Multipathing is a phenomenon in which the signal reaches the receiver via multiple paths, each having a unique modification. When combined, these alternations distort the received signal, leading to a potential corruption of the signal, especially in high-rate communication systems. To reduce the effect of multipathing, modern day telecommunication utilizes a combination of OFDM technology and MIMO system, ensuring a high speed connection while keeping the signal clean for easier processing. The content of this paper can be summarized as follows:

- Explain the theory and mathematical concepts used in a MIMO OFDM system.
- Present the program structure and analyzed contained code snippets.
- Evaluate the system under modified settings and conditions.

2 Theory

2.1 Overview

Orthogonal frequency-division multiplexing (OFDM) is a type of digital transmission used in digital modulation to encode digital (binary) data on multiple carrier frequencies. It transmits data across multiple frequencies that have orthogonal properties to each other.

A time-discrete sinusoidal signal is usually represented as a sinc function in the frequency domain. When waves of orthogonal frequency are used, it ensures that when one signal peaks in the frequency domain, the signals of others reach zero.

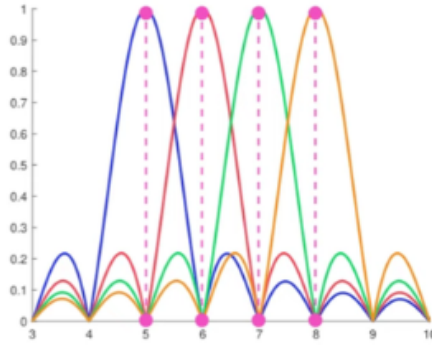


Figure 1: Orthogonal waves in the frequency domain

This approach helps to remove interference across tightly packed data in a limited bandwidth. We can experience an increase in data transmission rate up to n -times, with n being the number of subcarriers, compared to traditional, linear transmission.

Modern systems combine OFDM with MIMO system for cleaner data transmission with even higher data rate. A cyclic prefix can also be added to reduce signal overlapping due to delays and phase shifts.

We use the Zero-forcing method to combat interference from the MIMO system and clean up the signal. The process is applied to the received signal using 2D-interpolation with estimations based on a set of predetermined pilots symbols. QAM is a common and simple method to encode data for the OFDM algorithm. This simulation uses 16-QAM modulation scheme to encode an 8-bit character, with 2 constellations points for each character. Higher schemes such as 32-QAM or 64-QAM may be considered for higher bit-per-character.

2.2 Mathematical Concepts

Consider an OFDM matrix X of $N \times M$, where n represents the number of subcarriers of orthogonal frequencies and m represents the number of symbols for each subcarrier. If we consider one symbol is transmitted in one unit of time, it can also be stated that the column i ($1 \leq i \leq M$) of such matrix defines symbols transmitted in time i . To get the combined signal of every frequencies in the time segment i , we apply the inverse Discrete Fourier Transform (IDFT) to the i -th column. By appending M wave segments, we get a complete OFDM signal ready for transmission. The receiver side can reverse the process to reconstruct the original OFDM grid.

To compute the Discrete Fourier Transform (DFT), we use the DFT matrix, which is define as:

$$DFT(N) = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \omega^3 & \cdots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \omega^6 & \cdots & \omega^{2(N-1)} \\ 1 & \omega^3 & \omega^6 & \omega^9 & \cdots & \omega^{3(N-1)} \\ \cdots & \cdots & \cdots & \cdots & \ddots & \cdots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} & \omega^{3(N-1)} & \cdots & \omega^{(N-1)(N-1)} \end{bmatrix} \quad (1)$$

Where ω is given as:

$$\omega = \exp \frac{-2\pi i}{N} \quad (2)$$

From here, X in the frequency domain ($X(s)$) can be calculated as:

$$X(s) = DFT(N) * X \quad (3)$$

Evidently, we can inverse the DFT matrix to get the IDFT matrix to reverse the process:

$$X = IDFT * X(s) \quad (4)$$

When a signal is transmitted through a channel, it is prone to channel effects. With a $N \times M$ matrix, the effect on each symbol can be represented in mathematical form as:

$$X_{received} = H \odot X \quad (5)$$

Where H is the channel coefficient matrix:

$$H = \begin{bmatrix} H_{11} & H_{12} & \cdots & H_{1M} \\ H_{21} & H_{22} & \cdots & H_{2M} \\ \cdots & \cdots & \ddots & \cdots \\ H_{N1} & H_{N2} & \cdots & H_{NM} \end{bmatrix} \quad (6)$$

Through the use of a pilot symbol matrix P , we can estimate the channel coefficient matrix at specific points on the data:

$$\hat{H}_{pilot} = P_{receive} P^{-1} \quad (7)$$

Through the process of 2D-interpolation, the rest of the coefficient matrix can be estimated.

3 Simulation

The procedure of the simulation is visualized in the below diagram:

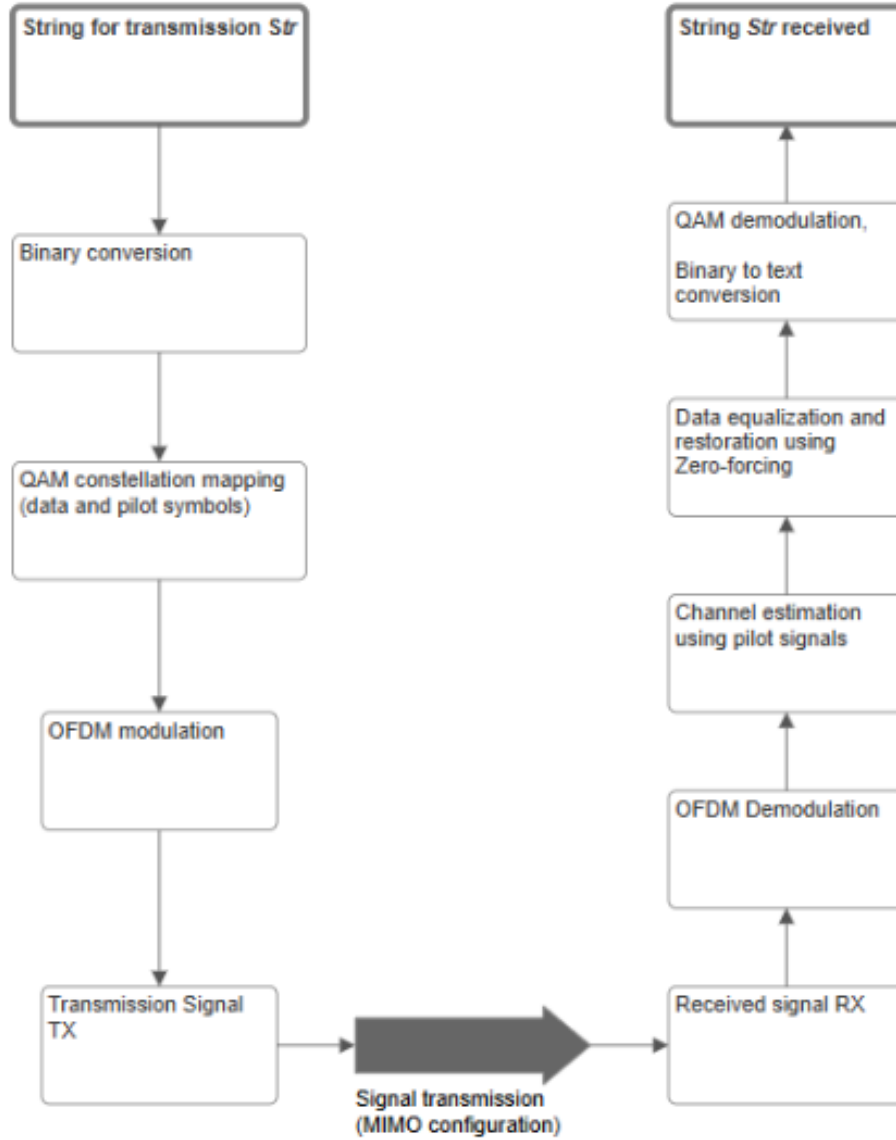


Figure 2: Block diagram of the simulation program

We have the features of the simulation system:

- 2 x 2 MIMO.

- 128 subcarriers, 20 symbols per subcarrier.
- 16-QAM modulation for 8-bit encoding.
- Zero-forcing equalizer.
- 20-symbol cycle prefix length

We use the built-in function *MIMOChan* from Matlab's communication toolbox to simulate noise and interference. The initial run is given the following parameters:

- 1.5 μ s delay
- 5 dB attenuation
- 20 Hz maximum doppler shift
- Ideal antenna independence

4 Evaluation

4.1 Default parameters

Consider the transmitted string to be $S_1 = "I \text{ love Mobile Communication}"$ and $S_2 = "abcdefghijklmnopqrstuvwxyz"$ for transmitter tx1 and tx2 respectively. Using the initial parameters give us the following result:

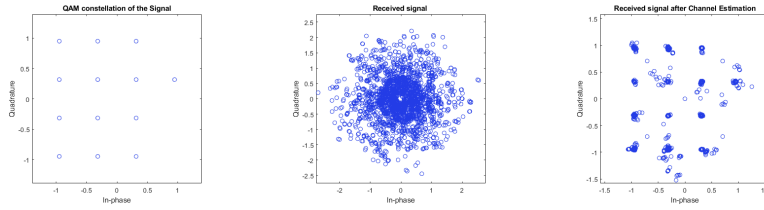


Figure 3: From left to right: The initial QAM constellation of the S_1 ; The received constellation of S_1 ; The constellation of S_1 after Equalization

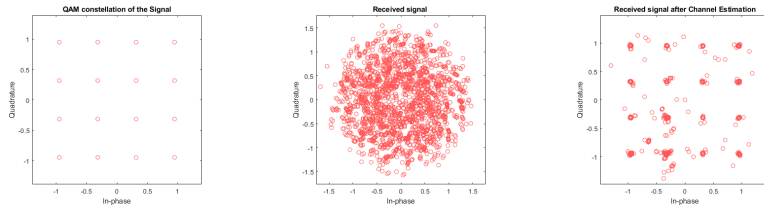


Figure 4: From left to right: The initial QAM constellation of the S_2 ; The received constellation of S_2 ; The constellation of S_2 after Equalization

We can see that the system perform as expected with the reconstruction of the data constellation. This is reflected in the output with very little corruption and a very low bit-to-error-ratio (BER) of 0.00625 and 0.0022321 for the respective data strings:

```
Port 1
BER: 0.00625
Data: I love Mobile Communication I lmf Mobile Communication I love Mobile Communication I love Mobile Communication I love Mobile Communication I love Mobile Communication I love Mobile C
Doppler frequency = 20
Port 2
BER: 0.0022321
Data: abodefghijklmnopqrstuvwxyz abodefghijklmnopqrstuvwxyz abodefghijklmnopqrstuvwxyz abodefghijklmnopqrstuvwxyz abodefghijklmnopqrstuvwxyz abodefghijklmnopqrstuvwxyz abodefghijklmnopqrstu
```

Figure 5: Output of S_1 (Port 1) and S_2 (Port 2)

4.2 Modification to the Cyclic Prefix

We adjust the length of the Cyclic Prefix (cp_length) to observe the impact of it on the performance of the system. For simplicity we only consider the performance of TX1 on the string S_1 .

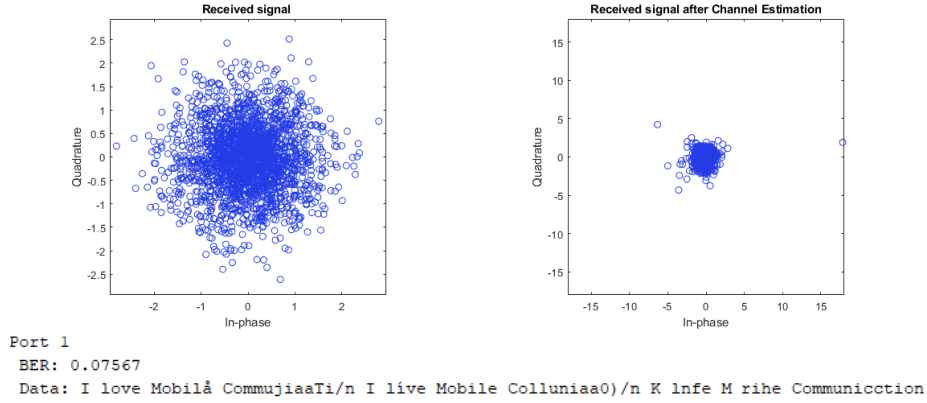


Figure 6: Without the Cyclic Prefix: The received constellation; The constellation after Equalization and the output of the data

To further realize the relationship between cp_length , the BER is plotted against the cp_length ranging from 0 to 100 symbols.

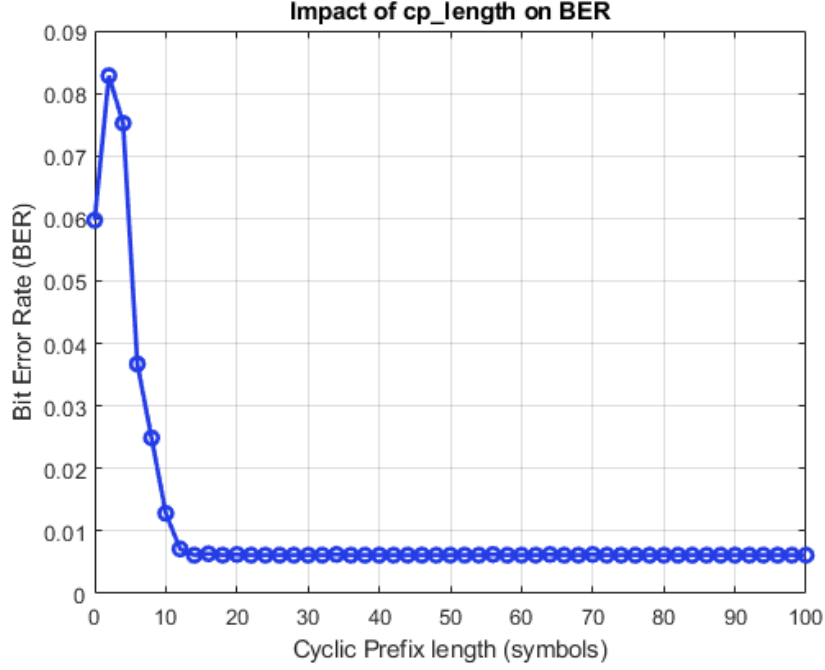


Figure 7: BER with every cp_length from 0 to 100 symbols

It is clear that without the cyclic prefix, overlapping causes a significant hike in BER (0.7567 compared to 0.00625 in the default setting). The error rate decrease sharply with every increase of the prefix length, eventually settle down at around 15 symbols of length. Any further increase in cp_length proves to be redundant.

4.3 Modification to the Doppler frequency

When the receiver is actively moving toward or away from the transmitter, communication suffers from the Doppler effect, which shifts the phase of the signal.

We set the Doppler frequency to 900 Hz, then compare the result with the original program, where the Doppler shift was 20 Hz.

This again interferes the constellation reconstruction process:

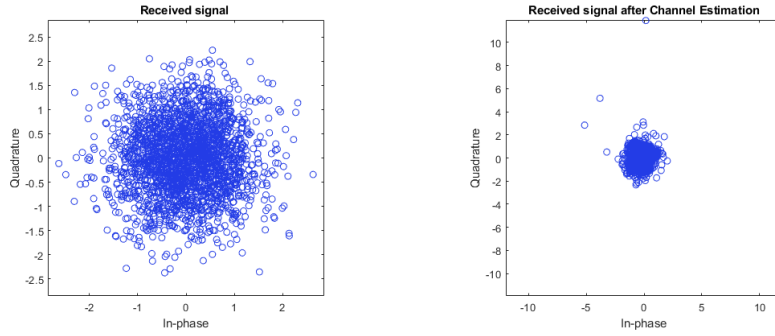


Figure 8: Doppler frequency at 700Hz: The received constellation; The constellation after Equalization

We can plot the changes in BER at Doppler frequency ranging from 0 Hz to 3 KHz:

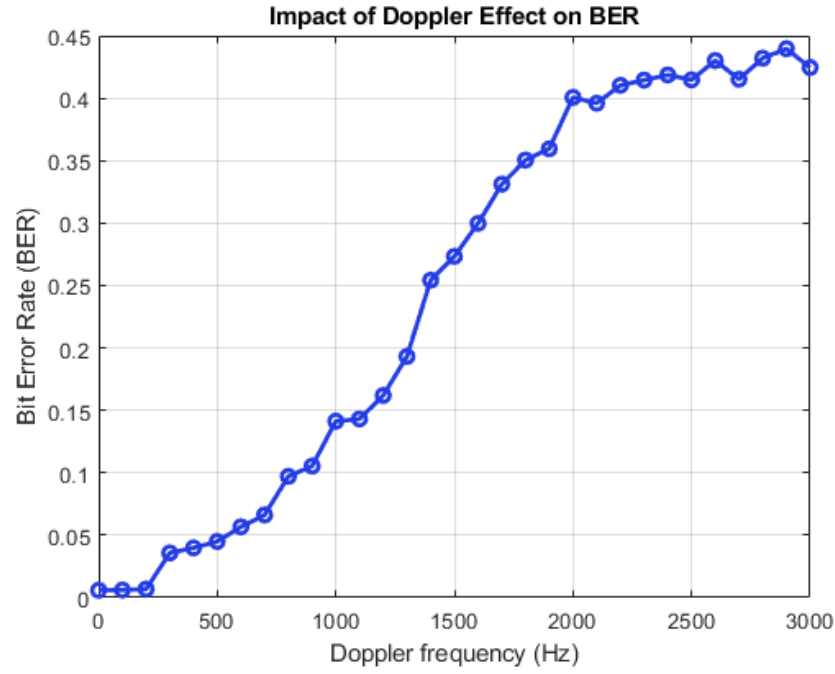


Figure 9: Doppler frequency at 700Hz: The received constellation; The constellation after Equalization

5 Conclusion and Future works

This paper illustrated the process of setting up a simulation for a 2x2 MIMO OFDM system and highlighted the mathematical concepts behind the applied algorithms. It also evaluated scenarios where key parameters were adjusted to prove the flexibility of the program.

In future researches, our group would like to increase the capability of the simulation by using more advanced MIMO configurations, modulation schemes and the possibility of Orthogonal frequency-division multiple access (OFDMA) implementation. For the current simulation, more in-depth analysis can be performed to realized the relationship between parameters, leading to discoveries that could potentially improve existing implementations.

6 References

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